Exercice 1

1/ y_i et y_r OPH sens $x \nearrow$

y_n ——— $\searrow$

2/ à la jonction :

$$y_i(0,t) + y_n(0,t) = y_r(0,t)$$

$$\rightarrow y_{0i} e^{j\omega_i t} + y_{0n} e^{j\omega_i t} = y_{0r} e^{j\omega_t}$$

$$\rightarrow \underbrace{(y_{0i} + y_{0r})\, e^{j(\omega_i - \omega_t)t}}_{\text{ça aussi}} = \underbrace{y_{0r}}$$

ça aussi $\Leftarrow$ indép de t

$$\rightarrow \omega_i = \omega_r$$

$$\rightarrow \omega_i = \omega_n = \omega_t =: \omega$$

3/ PFD sur la jonction de masse nulle :

$$0 = -T(0^-) + T(0^+)$$

$$\rightarrow \alpha(0^-) = \alpha(0^+)$$

$$\rightarrow \left.\frac{\partial y_i}{\partial x}\right|_0 + \left.\frac{\partial y_r}{\partial x}\right|_0 = \left.\frac{\partial y_t}{\partial x}\right|_0$$

$$-y_{0i}\, j k_i + y_{0r}\, j k_r = -y_{0t}\, j k_t \quad (1)$$

de +, $y_i(0) + y_r(0) = y_t(0)$

$$\rightarrow y_{0i} + y_{0r} = y_{0t} \quad (2)$$

$\rightarrow$ (1) + jk_t(2) :

$$y_{0i}(-jk_i + jk_t) + y_{0r}(jk_r + jk_t) = 0$$

$$\rightarrow r = \frac{y_{0r}}{y_{0i}} = \frac{k_t - k_i}{k_r + k_t} = \frac{\frac{1}{c_2} - \frac{1}{c_1}}{\frac{1}{c_2} + \frac{1}{c_1}}$$

$$r = \frac{c_1 - c_2}{c_1 + c_2}$$

$(1) - jk_r\,(2)$:

$$y_{0i}\left(-jk_i - jk_r\right) = y_{0t}\left(-jk_t - jk_1\right)$$

$$\rightarrow t = \frac{y_{0t}}{y_{0i}} = \frac{k_i + k_r}{k_t + k_r} = \frac{\frac{1}{c_1} + \frac{1}{c_1}}{\frac{1}{c_2} + \frac{1}{c_1}}$$

$$t = \frac{2c_2}{c_1 + c_2}$$

Exercice 2

1/ On reprend le PFD du cours avec en plus la force de Laplace

$$\mu \, dx \frac{\partial^2 z}{\partial t^2} \vec{u}_z = \vec{T}(x+dx) - \vec{T}(x) + i \, dx \, B \vec{u}_z$$

$$\rightarrow \mu \frac{\partial^2 z}{\partial t^2} = T \frac{\partial \alpha}{\partial x} + i_0 \cos(\omega t) B_0 \sin\left(\frac{\pi x}{L}\right)$$

$$\rightarrow \frac{\partial^2 z}{\partial t^2} - c^2 \frac{\partial^2 z}{\partial x^2} = \frac{i_0 B_0}{\mu} \cos(\omega t) \sin \frac{\pi x}{L}$$

avec $c = \sqrt{\frac{T}{\mu}}$

2/ Corde accrochée → onde stationnaire qui vibre à la fréquence d'excitation.

3/ $-z_0 \omega^2 \sin\left(\frac{\pi x}{L}\right) \cos(\omega t)$

$+c^2 \left(\frac{\pi}{L}\right)^2 z_0 \sin\left(\frac{\pi x}{L}\right) \cos(\omega t)$

$= \frac{i_0 B_0}{\mu} \cos(\omega t) \sin\left(\frac{\pi x}{L}\right)$

$$\rightarrow Z_0 = \frac{iB_0/\mu}{c^2 \frac{\pi^2}{L^2} - \omega^2}$$

quand $\omega \rightarrow \frac{\pi c}{L}$, Z_0 diverge.

On ne peut alors plus supposer Z et α petits ni négliger les frottements avec l'air

Exercice 3

1/ $\underline{u}_i = \underline{u}_{oi}\, e^{j(\omega t - k_i x)}$

$\underline{u}_r = \underline{u}_{or}\, e^{j(\omega t + k_r x)}$

$\underline{u}_t = \underline{u}_{ot}\, e^{j(\omega t - k_t x)}$

2/ $$\begin{cases} \underline{u}_i(0,t) + \underline{u}_r(0,t) = \underline{u}_t(0,t) \\ \dfrac{\underline{u}_i(0,t)}{Z_1} - \dfrac{\underline{u}_r(0,t)}{Z_1} = \dfrac{\underline{u}_t(0,t)}{Z_2} \end{cases}$$

3/ $2\underline{u}_{io} = \underline{u}_{ot}\left(1 + \dfrac{Z_1}{Z_2}\right)$

$\rightarrow t = \dfrac{\underline{u}_{ot}}{\underline{u}_{oi}} = \dfrac{2Z_2}{Z_1 + Z_2}$

$\underline{u}_{oi}\left(1 - \dfrac{Z_2}{Z_1}\right) + \underline{u}_{or}\left(1 + \dfrac{Z_2}{Z_1}\right)$

$= 0$

$$\rightarrow \quad r = \frac{\underline{u}_{0r}}{\underline{u}_{0i}} = \frac{\frac{Z_2}{Z_1} - 1}{1 + \frac{Z_2}{Z_1}} = \frac{Z_2 - Z_1}{Z_1 + Z_2}$$

4/ $$R = \left| \frac{\langle u_r \, i_r \rangle}{\langle u_i \, i_i \rangle} \right| = \left| \frac{\langle Z_1 u_r^2 \rangle}{\langle Z_1 u_i^2 \rangle} \right|$$

$$= \left| \frac{\langle r^2 u_i^2 \rangle}{\langle u_i^2 \rangle} \right|$$

$$= |r^2| = \left(\frac{Z_2 - Z_1}{Z_1 + Z_2} \right)^2$$

de m̂ $$T = \left| \frac{\langle u_r \, i_r \rangle}{\langle u_i \, i_i \rangle} \right|$$

$$= \left| \frac{\langle Z_2 u_r^2 \rangle}{\langle Z_1 \, u_i^2 \rangle} \right|$$

$$= \left| \frac{Z_2}{Z_1} t^2 \right| = \frac{2 Z_1 Z_2}{(Z_1 + Z_2)^2}$$

$R + T = 1$

Exercice 4

1/ $\underline{u}(x,t) = \underline{u}_0^+ e^{j(\omega t - k x)}$

$+ \underline{u}_0^- e^{j(\omega t + k x)}$

$\rightarrow \underline{i}(x,t) = \dfrac{\underline{u}_0^+}{Z_c} e^{j(\omega t - k x)}$

$- \dfrac{\underline{u}_0^-}{Z_c} e^{j(\omega t + k x)}$

$\underline{u}(0,t) = E_0 e^{j\omega t}$

$\rightarrow \underline{u}_0^+ e^{j\omega t} + \underline{u}_0^- e^{j\omega t} = E_0 e^{j\omega t}$

$\rightarrow \underline{u}_0^+ + \underline{u}_0^- = E_0 \quad (1)$

$\underline{i}(L,t) = 0$

$\rightarrow \dfrac{\underline{u}_0^+}{Z_c} e^{j(\omega t - kL)} - \dfrac{\underline{u}_0^-}{Z_c} e^{j(\omega t + kL)} = 0$

$$\rightarrow \underline{u}_0^+ e^{-jkL} - \underline{u}_0^- e^{jkL} = 0 \quad (2)$$

$(1) \cdot e^{jkL} + (2)$

$$\rightarrow \underline{u}_0^+ \overbrace{\left(e^{jkL} + e^{-jkL}\right)}^{2\cos(kL)}$$

$$+ \underline{u}_0^- \left(\cancel{e^{jkL} - e^{jkL}}\right)$$

$$= E_0 e^{jkL}$$

$$\rightarrow \underline{u}_0^+ = \frac{E_0}{2\cos(kL)} e^{jkL}$$

$(1) e^{-jkL} - (2)$:

$$\underline{u}_0^+ \left(\cancel{e^{-jkL} - e^{-jkL}}\right)$$

$$+ \underline{u}_0^- \underbrace{\left(e^{-jkL} + e^{jkL}\right)}_{2j\cos(kL)}$$

$$= E_0 e^{-jkL}$$

$$\rightarrow \underline{U}_0^- = \frac{E_0}{2\cos(kL)} e^{-jkL}$$

$$\rightarrow \underline{U}(x,t) = \frac{E_0}{2\cos(kL)} \left(e^{j(\omega t - kx + kL)} + e^{j(\omega t + kx - kL)} \right)$$

$$= \frac{E_0}{2\cos(kl)} e^{j\omega t} \cdot 2\cos(kx - kL)$$

$$\underline{i}(x,t) = \frac{E_0}{2\cos(kL) Z_0} \left(e^{j(\omega t - kx + kL)} - e^{j(\omega t + kx - kL)} \right)$$

$$= \frac{E_0}{2\cos(kL) Z_0} e^{j\omega t} \cdot 2j\sin(kL - kx)$$

$$\rightarrow \begin{cases} u(x,t) = \dfrac{E_0}{\cos(kL)} \cos(\omega t) \cos(kx - kL) \\ i(x,t) = \dfrac{E_0}{\cos(kL) Z_0} \cos\left(\omega t + \frac{\pi}{2}\right) \sin(kL - kx) \end{cases}$$

2/ $kL = \frac{\pi}{2} + p\pi \quad ; p \in \mathbb{N}$

$\Rightarrow u \rightarrow +\infty$

(résonnance)

Exercice 5

1/ à cause des CL

2/ PFD → $\rho \dfrac{\partial \vec{v}}{\partial t} = -\overrightarrow{grad}\, P$

$$= -A_0 \cos(\omega t)\, k \sin(kx)\, \vec{u}_x$$

$$\rightarrow \vec{v} = -\frac{A_0 k}{\rho \omega} \sin(\omega t) \sin(kx)\, \vec{u}_x$$

non

3/ $\begin{cases} \vec{v}(x=0) = \vec{0} \\ P(x=l) + P_0 = P_0 \end{cases}$

4/ $\cos(kl) = 0$

$$\rightarrow kl = \frac{\pi}{2} + n\pi$$

$$\rightarrow \omega = \frac{c}{l}\left(\frac{\pi}{2} + n\pi\right)$$

5/ Fondamentale : $\omega = \frac{c}{l}\frac{\pi}{2}$ Soit

2× plus grave qu'une flute

(1 octave plus grave)

6/ Equivalent à une clarinette de 37,5 cm

: $f = \frac{\omega}{2\pi} = \frac{c}{8l} = \frac{340}{8 \times 0,375} = 113 Hz$

Exercice 6

1/ $\underline{E}$ OS PH dans le sens $r \nearrow$ et $\searrow$

2/
$$\rho \frac{\partial \vec{\underline{v}}}{\partial t} = -\overrightarrow{\text{grad}}\ \underline{P}$$
$$= -\left(-\frac{A}{r^2} e^{j(\omega_1 t - k_1 r)} - \frac{A}{r} j k_1 e^{j(\omega_1 t + k_1 r)} - \frac{B}{r^2} e^{j(\omega_2 t - k_2 r)} + \frac{B}{r} j k_2 e^{j(\omega_2 t + k_2 r)} \right) \vec{u}_r$$

$$\Rightarrow \vec{\underline{v}} = \frac{A}{\rho\omega_1}\left(\frac{1}{r^2} + j\frac{k_1}{r}\right) e^{j(\omega_1 t - k_1 r)} \vec{u}_r + \frac{B}{\rho\omega_2}\left(\frac{1}{r^2} - j\frac{k_2}{r}\right) e^{j(\omega_2 t + k_2 r)} \vec{u}_r$$

3/ $\vec{\underline{v}}(R) = \vec{0}$

$$\rightarrow \frac{A}{\rho\omega_1}\left(\frac{1}{R^2} + j\frac{k_1}{R}\right) e^{j(\omega_1 t - k_1 R)}$$

$$= \frac{B}{\rho\omega_2}\left(j\frac{k_2}{R} - \frac{1}{r^2}\right) e^{j(\omega_2 t + k_2 R)}$$

$$\rightarrow \begin{cases} \omega_1 = \omega_2 \\ A\left(\frac{1}{R^2} + j\frac{k}{R}\right) e^{-jkR} = B\left(\frac{jk}{R} - \frac{1}{R^2}\right) e^{jkR} \end{cases}$$

4/ La pression ne diverge pas en O

$$\rightarrow A = -B$$

$$\rightarrow A\left(\frac{1}{R^2} + j\frac{k}{R}\right) e^{jkR} = A\left(\frac{1}{R^2} - j\frac{k}{R}\right) e^{-jkR}$$

$$\rightarrow \frac{\frac{1}{R^2} + j\frac{k}{R}}{\frac{1}{R^2} - j\frac{k}{R}} = e^{-2jkR}$$

$$\rightarrow \quad -2kR = \arg \frac{\frac{1}{R^2} + j\cdot\frac{k}{R}}{\frac{1}{R^2} - j\cdot\frac{k}{R}}$$

$$= \arg \frac{\frac{1}{R^4} - \frac{k^2}{R^2} + 2j\frac{k}{R^3}}{\cdots \in \mathbb{R}}$$

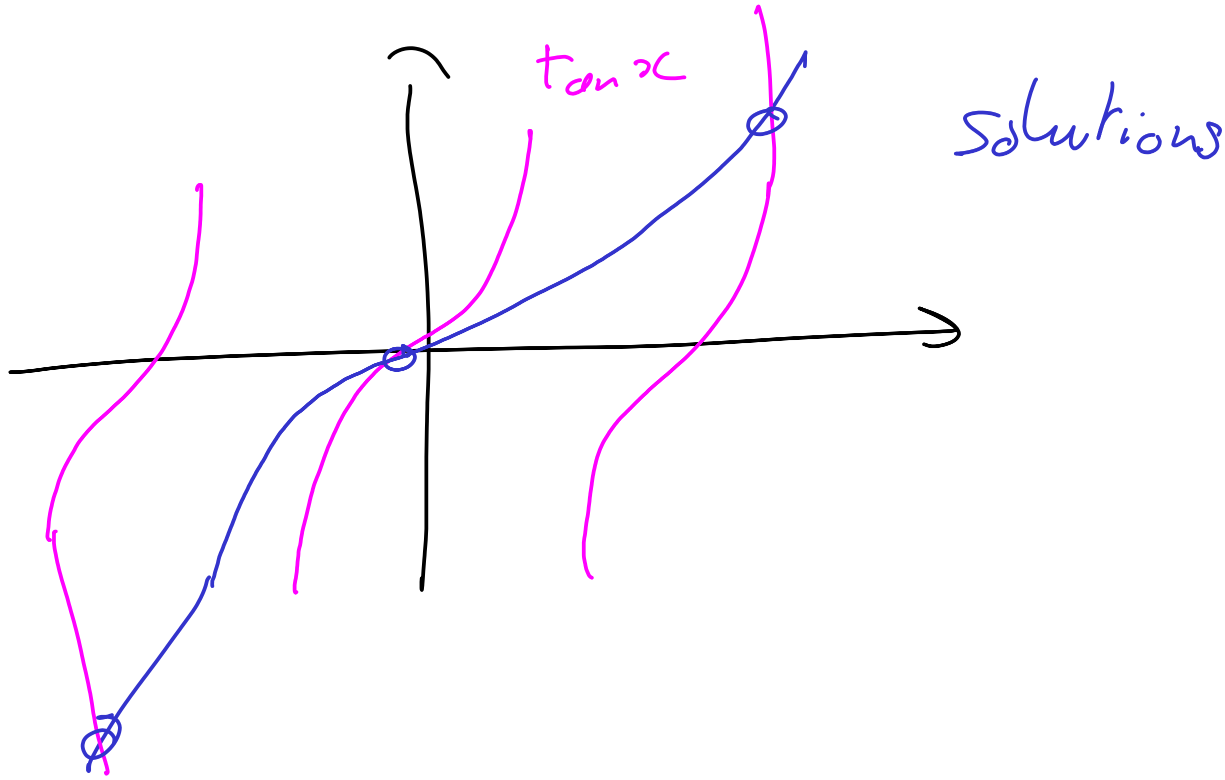
$$= \arctan \frac{2k/R^3}{\frac{1}{R^4} - \frac{k^2}{R^2}}$$

$$-2kR = \arctan \frac{2kR}{1 - k^2R^2}$$

$$\rightarrow \tan(2kR) = \frac{2kR}{1-k^2R^2}$$

Si $x = 2kR$

$$\rightarrow \quad \tan x = \frac{x}{1 - \frac{x^2}{4}}$$


tan x
solutions

Exercice 7

1/ Cours $\rightarrow \vec{\Delta} \vec{E} - \frac{1}{c^2} \frac{\partial^2 \vec{E}}{\partial t^2} = \vec{0}$

2/ $f''(x) g(t) - \frac{1}{c^2} f(x) g''(t) = 0$

$\rightarrow \frac{f''(x)}{f(x)} = \frac{1}{c^2} \frac{g''(t)}{g(t)}$

$\rightarrow = \text{cte} =: \alpha$

$\rightarrow f'' = \alpha f$ et $g'' = c^2 g$

3/ $\vec{E}(x=0) = \vec{E}(x=a) = \vec{0}$

4/ $\cdot\ \alpha > 0$

$\rightarrow f(x) = Ae^{\alpha x} + Be^{-\alpha x}$

$f(0) = 0 \rightarrow A + B = 0$

$f(a) = 0 \rightarrow Ae^{\alpha a} + Be^{-\alpha a} = 0$

$\rightarrow \quad A e^{\alpha a} = A e^{-\alpha a}$

$\rightarrow \quad A = 0$

$\rightarrow$ pas d'onde

- $\alpha = 0 \quad \rightarrow \quad f(x) = (A + Bx)\, e^{\alpha x}$

idem, les CL ne permettent pas d'onde

- $\alpha < 0$

$\rightarrow \quad f(x) = A \cos(k x) + B \sin(kx)$

avec $k = \sqrt{-\alpha}$

$f(0) = 0 \rightarrow \quad B = 0$

$f(a) = 0 \rightarrow \quad \cos(ka) = 0$

$\rightarrow \quad ka = \frac{\pi}{2} + n\pi$

$\rightarrow \quad k = \dfrac{\frac{\pi}{2} + n\pi}{a}$

5/ $g(t) = A\cos(\omega t + \varphi)$

avec $\omega = kc$

$\rightarrow \vec{E} = E_0 \cos(\omega t + \varphi)\cos(kx)\vec{u_y}$

6/ Corde de Melde

7/ $\vec{rot}\,\vec{E} = -\dfrac{\partial \vec{B}}{\partial t}$

$$\begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix} \wedge \begin{pmatrix} 0 \\ E(x,t) \\ 0 \end{pmatrix} = \frac{\partial E}{\partial x}\vec{u_z}$$

$$= -E_0 k \cos(\omega t + \varphi)\sin(kx)\vec{u_z}$$

$\rightarrow \vec{B} = E_0 \dfrac{k}{c_0} \sin(\omega t + \varphi)\sin(kx)\vec{u_z}$

Les noeuds de $\vec{B}$ sont des ventres de $\vec{E}$ et réciproquement.